RANKERS BATCH

NEET

Botany

Cell cycle and cell division

one shot

TARUN SIR

Basics of cell cycle

– Every organism Derive From Single cell (From Zygote Division), Cell Division & Growth is feature of cell (Cell Growth → Divide into two Daughter)

Cell cycle → Coordinate Event → Cell Growth, DNA Double Divide & Distribute → intact genome two Daughter cell (Specific phase)

Feature of cell cycle →
- Coordinated ✓
- → Genetic control ✓
- → one phase DNA Double, one phase Genome distribute into two cell

(Eukaryotic cell) ↓ Cell cycle → Phase (two phase)

① M phase – cell Division
- 5% time
- Karyo Kinesis – nucleus Divide
- Cyto Kinesis → Cytoplasm

② Interphase → B/w two Sucessive M-phase
- 95% time of cell cycle
- Resting phase → cell prepare For Division

→ $G_1 + S + G_2$ = Interphase

G_1 → Gap 1 B/w M & S

G_2 → Gap-2 (Gap B/w S & M-phase)

Topics – ① Time

→ Yeast cell – 90 min Human cell – 24 hr (Culture)

② time (vary → two cell of same organism – two cell of Different organism]

G_1 (8-12 hr) > S (6-8 hr) > G_2 (3-4 hr) > M (1 hr) → cell Divide 24 hr

ⅡChanges in phase

G_1 → Post Mitotic, Gap-1 (transcription, translation) Growth in Cytoplasm / Cell occur

→ three option (G1 cell) — ① Cell Become permanent ② Enter into S-phase ③ Exit G_1 → Enter G_0 (quiscent Phase) – cell living, metabolically Active (not Proliferate)

S – Synthetic (DNA Double in nucleus) (Histone synthesis) [Centriole Double in cytoplasm – Animal cell

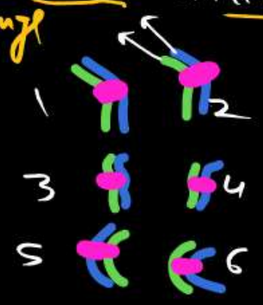
G_2 → Gap-2 – cell prepare For Division (Pre mitotic) (Mitochondria Plastid Double in G_2)
Protein synthesis → tubulin → Give Microtubule

III Amount change Ploidy change

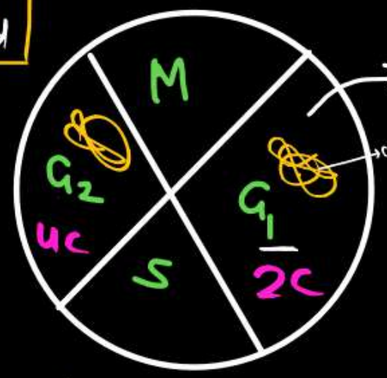
→ Cell cycle → number of chromosome – not change

Chromosome = 6
Chromatid = 12
Centromere = 6

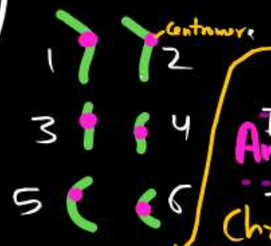
1 chromosome
2 chromatid



Diploid cell



(Example)
→ total chromosome = 6
Chromatid → 6
Amount → 2c (Amount of DNA of chromatin)

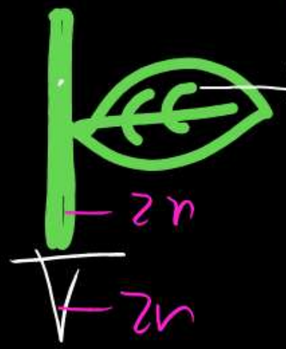


G_1 phase → chromatin visible
if chromosome visible → Single chromatid

	G_1	S	G_2	M
DNA Amount	2c	2c–4c	4c	4c → 2c → 2c
No Chromosome	6	6	6	6
Chromatid	6	Change	12	6 } Each Daughter cell 6

(Amount of DNA change)
→ Amount change in phase → S phase (2c → 4c), M-phase (4c → 2c → 2c)
→ Number of chromatid change → G_1 phase – 6 chromatid | 6 chromosome
G_2 phase → 12 chromatid | 6 chromosome
→ Number of chromosome change → NO

Diploid → G_1 – 2c ✓
G_2 = 4c ✓
haploid → c (Amount of DNA)
if Haploid Divide ☆
G_1 – c G_2 = 2c



(Vegetative cell | Somatic cell)
→ Leaf cell → 24 chromosome
(Diploid cell)

→ Chromosome in Root, Stem = 24 chromosome
→ Chromosome of in G_1 → 24
Any cell
→ Chromatid in G_1 → 24
G_2 → 48
→ Chromosome in Pollen mother cell (2n) → 24
→ Amount of DNA → PMC (2n) → G_1 & G_2 → 4c
Pollen → c
Pollen → n → 12
G_1 → 2c

Mitosis (M-phase)

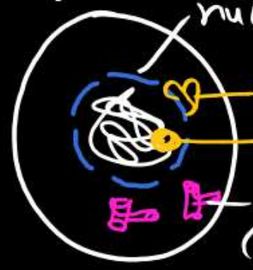
→ Karyokinesis (P M A T) → longest – Prophase, Shortest → Anaphase
- Prophase
- Metaphase
- Anaphase
- telophase

→ Cytokinesis (Cytoplasm Divide) – Reorganization of all component (nucleus & cytoplasm)

→ Most Dramatic phase

→ Equational Division (No. of chromosome – Parent & Daughter cell same)

Prophase



Chromatin Double (S-phase) → Still interwined (उलझी) → Untangling of Chromatin

– Chromatin condense (not Fully) → Chromosome visible – Early Prophase

– Centrosome move opposite pole
– Spindle Form – Mid Prophase

Centrosome + Microtubule = Aster
two Aster + spindle = spindle Apparatus

– ER, Nucleolus, Golgi – Disappear
– Nuclear Membrane start to Disintegrate – late Prophase

Metaphase →

First Event – nuclear Membrane Break (completely)

[Chromosome come in cytoplasm
– Chromosome Fully condensed – Study of Morphology possible]

→ Spindle Fibre attach on Kinetochore [Disc shape Proteinaceous present on Centromere]

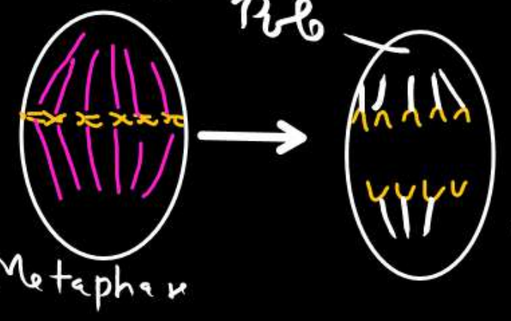
in Such a manner → 1 chromosome = 2 spindle Fibre Attach) → one pole, opposite pole

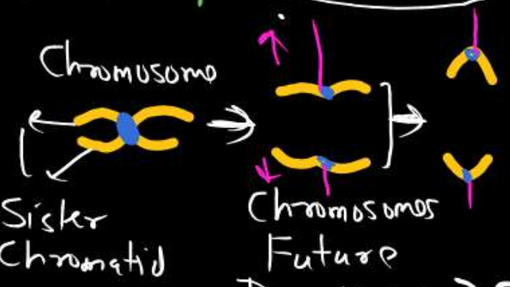
⇒ Chromosome → come in centre / Equator of spindle

Plane of Alignment of chromosome at Equator = Metaphasic plate

Anaphase

→ All chromosome – split Simultaneously → Sister chromatid Seperate





→ Sister chromatid is now Daughter Chromosome, Move opposite pole

→ Centromere Directed towards pole & Arms trailing Behind

Telophase

Chromosome – decondense (no longer individual chromosome) → Identity lost

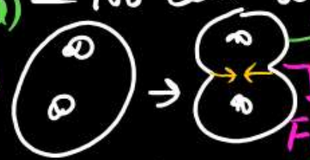
– Nuclear Membrane, ER, Golgi

Nucleolus → Reappear

Cytokinesis – Cytoplasm & organelles – Distribute in Daughter cell

(Animal cell) – No cell wall

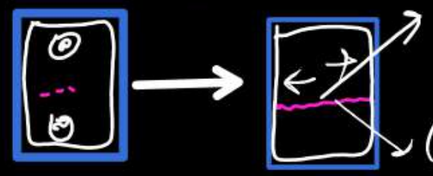
(a) Cleavage Method →



Furrow → deepens → Meet in centre

(Plants → Rigid cell wall)

(b) Cell Plate method → Plants → Middle lamella deposit (inner to outer)



Significance (Mitosis)

→ (a) Genetically identical cells

(b) nucleus/cytoplasm Ratio Restore

(c) Growth, Repair

(d) occur in Diploid cells in Animals

(Exception – Social insect – drone Honey bee) → Mitosis in haploid

(e) occur in plants → Diploid & Haploid (lower plants)

Location → Gut, Epidermis, Blood cells, Meristems

Growth – n/c dec)

Cell division – n/c inc

APC – Anaphase promoting complex

→ Apc Functional in

Digest By APC → Cohesin

If APC not Functional → Sister Chromatid Remain Attach (Polyteny)

Colchicine

→ (Chemical

→ Spindle Formation stop

→ Sister chromatid seperate → But Remain in same cell

Chromosome Double (Polyploidy)

Meiosis

→ occur in Diploid cell
→ Why meiosis → It produce haploid stage

→ S-phase - one time
Cell Divide → two time

Stages

Meiosis I

$G_1 \to S \to G_2$

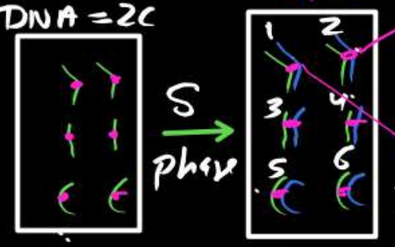
- Prophase - I
- Metaphase - I
- Anaphase - I
- Telophase - I

Interkinesis

Meiosis II

- P - II
- M - II
- A - II
- T - II

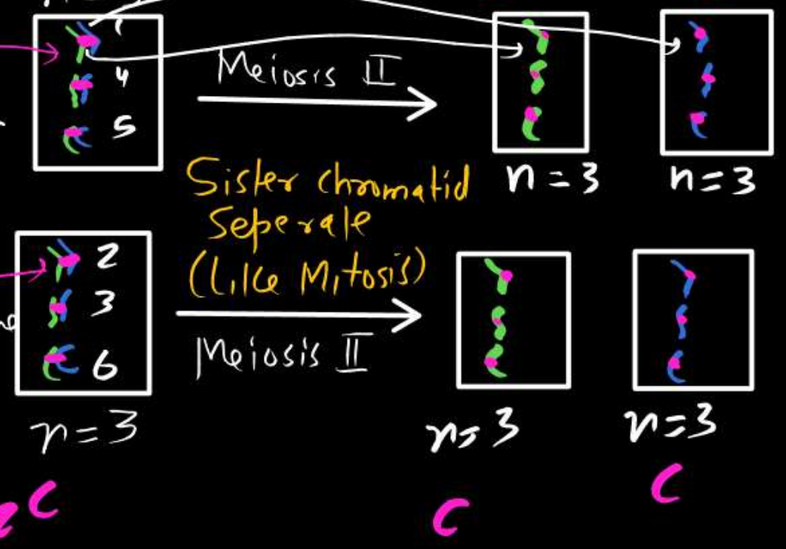
Amount of DNA = 2C



G_1 phase
Chromosome = 6
3 pair
(2n = 6)

G_2
2n = 6
4C

Meiosis I → Reduction (Chromosome no. half)



Meiosis (Reductional)

Feature

→ It give variation - Exchange of Genetic material (B/w chromosome)
→ - Pairing of chromosome occur
→ 4 haploid Daughter cell Form
- One cycle of DNA Replication, two cell Division
→ Mitosis → Equational Division
→ Meiosis → Reductional Division (chromosome half)

In Meiosis I - Changes → Chromosome number half (Chromosome Pair seperate) (Reduction)

In Meiosis II - Changes → chromosome number constant, Amount half (Equational)

Meiosis I

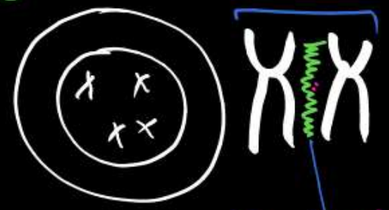
Prophase I
- More complex, longer - Divided into phases on the Basis of → Behaviour of chromosome
(Compare prophase - Mitosis)

L Z P D D

(A) **leptotene** → Chromatin condense
→ Condensation occur throughout

Chromosome visible (leptotene)

Zygotene → Pairing of chromosome occur →
Process of pairing → Synapsis

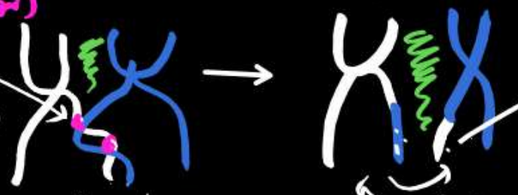


Paired chromosome is called — Bivalent (tetrad)
(Pair)

Synaptonemal complex → help in pairing
(visible in EM)

Human cell = 23 Bivalent or 23 tetrad

Pachytene - Bivalent More clearly visible - Pachytene
- Exchange of Genetic Material → B/w - Non-sister chromatid
(Crossing over)
- Recombination nodule
- Enzyme - Recombinase
- Result is
↳ Variation
(Exchange of Genetic Material)



non sister chromatid

Exchange of genetic material (Recombination)

Diplotene →
- Dissolution of → Synaptonemal complex
- chromosome start to move Away (Some occyte Arrest in this stage)
- Chiasmata visible

Cross like structure

Sister chromatids
non-sister chromatid

DIAKINESIS → terminilization of chiasmata
- Nuclear Membrane, ER, Golgi, nucleolus (Disappear)
- Chromosome Fully condensed
(This phase is transition to Metap

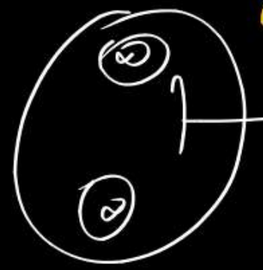
Metaphase I

Chromosome pair → spindle Attach From opposite pole
1 pair = 2 spindle Fibre
(Metaphasic plate - Pair of chromo

Anaphase I — Pair seperate

One chromosome From Each pair (move to opposite pole)

Telophase I → two pole → chromosome decondense → chromatin



→ two haploid nuclei From

(ER, Golgi, nucleolus, nuclear Membrane Appear Again)

Mitosis

Interphase

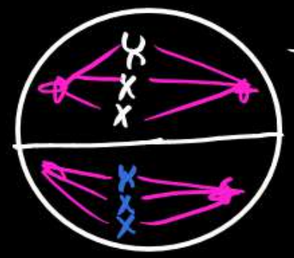
- Gap B/w two Mitosis
- → DNA Replication occur in S-phase

Meiosis-II (like Mitosis)

Prophase II →

- Chromatin condense
- ER, Golgi, nucleolus Disappear
- Nuclear Membrane Break

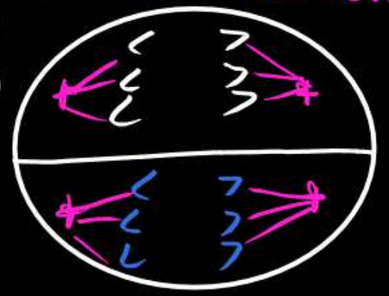
Metaphase II →



- Each chromosome have two spindle Fibre - opposite pole

Anaphase II → Sister Chromatid separate

(APC Functional)

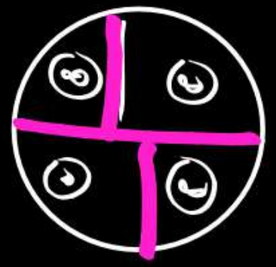


(Meiosis)

→ Inter Kinesis → Gap B/w M-I & M-II

- No DNA Replication
- Centriole Double
- DNA not come in complete Dispersed state of Chromatin

Telophase



→ 4 haplod nuclei form

→ Cyto Kinesis → 4 haploid cells (3) from

Significance → Meiosis → Chromosome half (Haploid stage Form) (But constant in species)

→ Variation By - Crossing over → Helpful in Evolution